Even Sum

PLATFORM	DIFFICULTY	DATE
Platform: CodeChef	DIV 3	Solved: June 3, 2026

Problem

Remove exactly one element such that the sum of the remaining elements is even.

Key Observation

Let total sum be:

$$S = A_1 + A_2 + \cdots + A_N$$

S=A1+A2+⋯+AN

After removing Ai:

$$\text{remaining sum} = S - A_i$$

remaining sum=S−Ai

For this to be even:

$$S - A_i \equiv 0 \pmod 2$$

S−Ai≡0(mod2)

So S and Ai must have the same parity.

Algorithm

- Calculate total sum.
- If sum is even:
  - Need at least one even element.
- If sum is odd:
  - Need at least one odd element.

TC & SC

Time Complexity : O(N)

Space Complexity : O(1)